A Finite Mathematical Framework Correlating Cognitive Behavioral Optimization, Polymathic Anchoring, and QML Neuromodulation in Schizoaffective Disorder

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April 2026

Abstract

We introduce a finite deterministic mathematical framework to model the complex neural correlates of Schizoaffective Disorder under the influence of pharmacological antipsychotics, limited <4.5% ABV beer intake, nicotinic (nAChR) self-medication, and polymathic physics computation. Traditional approaches treat substance use unconditionally; here, we precisely quantify boundaries where mild reward circuits activate without destabilizing prefrontal gating via mesolimbic dopamine volatility. We demonstrate mathematically that intense polymathic cognition acts as a structured dopamine sink, anchoring striatal hyperactivity into deterministic functional tracts. Furthermore, we outline the integration of Quantum Machine Learning (QML) optimized transcranial neuromodulation alongside Hebbian amplification and Cortical Control indices to predict and minimize clinical relapse probability while maximizing psychological efficiency.

1. Dopamine Volatility & Neuropharmacological Anchoring

The probability density function integrates beer (B), nicotine (N), meds (M), and polymath cognition (P). Polymathic effort P serves to sink volatile dopamine σ_d^2:

$$\sigma_d^2 = \max\left(0.1, \sigma_{base}^2 + 1.5(0.35B + 0.4N) - 1.2M - 0.6P\right)$$

This variance dictates the probability density function (PDF) of symptom severit

$$P(x) = \frac{1}{\sigma_d\sqrt{2\pi}} e^{-\frac{(x-\mu_{symp})^2}{2\sigma_d^2}}$$

2. Prefrontal Cortical Control Index (C_{pfc})

Executive oversight correlates prefrontal efficiency (A_{pfc}) over limbic hyperactiv

$$A_{pfc} = \max(0.1, 1.0 - 0.8(0.35B) + 1.2(0.4N) + 0.5M + 1.2P)$$

$$C_{pfc} = \min\left(1.0, \frac{A_{pfc}}{\max(0.1, \frac{1}{2}(A_{amy} + A_{hip}))}\right)$$

3. Hebbian Amplification Optimization for CBT

Neuroplastic efficiency for cognitive framing is maximized when prefrontal cont prevails over systemic variance:

$$H(\%) = \min\left(100.0, \max\left(0.0, \left[\frac{A_{pfc}}{\sigma_d^2}\right] \times 45.0\right)\right)$$

4. Minimal Relapse Probability Derivation

The deterministic probability of clinical relapse (P_{rel}) emerges as:

$$P_{rel} = 100.0 - 80.0(C_{pfc}) + 15.0(\sigma_d^2) - 10.0(M) - \Delta_{CBT}$$

Where $\Delta_{CBT} = 15.0$ iff the CBT Optimization Score ≥ 70.

5. QML Neuromodulation Trajectories (TMS)

Quantum state measurements of entangled subsystem proxies are iteratively op
to derive the exact therapeutic frequency for transcranial magnetic stimulation

$$F_{TMS} = 10.0 + 8.5 \cdot \sigma_d^2 \quad \text{(Hz)}$$

Coupled with structured polymathic activity, the intervention yield projects as:

$$\gamma_{yield}(\%) = \min(99.9, \max(0, 65.0 - 30.0B_{imp} + 15.0N_{imp} + 45.0M + 20.0$$